

Milan Short-Term vs Long-Term Rental Market

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1. Executive summary

Research Question: Which Milan neighborhoods generate the highest short-term rental premium over long-term rent, and what neighborhood characteristics may account for this variation?

Approach:

- Merged Airbnb listings data (22,948 listings) with long-term rent data (31 macro-zones)
- Computed neighbourhood-level premium metric for 87 neighbourhoods using consistent time window (June 2025–March 2026)
- Built interactive decision-support dashboard for landlords to evaluate rental strategies by location and apartment size
- Used Claude Code throughout for data cleaning, analysis, visualization, dashboard development, and report drafting

Key Findings:

- Short-term rental premiums range from +40% to +141%, with Duomo ranking highest
- High premiums appear in both central tourist areas (Duomo, Brera) and peripheral neighbourhoods (S. Siro, Baggio)
- Rent level alone explains only 5% of premium variation ($R^2 = 0.05$, $p = 0.078$)
- Geographic distribution is scattered across the city, not concentrated in a central tourist corridor
- Two paths to high premiums: high tourist demand with premium nightly rates, or low baseline rent with moderate Airbnb revenue

Recommendations:

1. Evaluate at neighbourhood level – citywide averages mask significant spatial variation
2. Consider peripheral areas – some outer neighbourhoods with low baseline rents show strong short-term rental potential
3. Interpret premiums cautiously – comparison reflects gross revenue, not net profit after operating costs, vacancy, and regulation

Bottom Line: Short-term rental profitability in Milan depends less on headline rent levels and more on localized market conditions. The dashboard provides landlords with neighbourhood-specific estimates to support informed rental decisions.

2. Introduction

2.1 Context

Milan receives over 10 million visitors a year for business, fashion, trade fairs, and tourism. That visitor volume has created a large market for short-term accommodation, and Airbnb has given property owners an alternative to traditional long-term leasing. But the financial trade-off between the two approaches is hard to evaluate without neighborhood-level data.

Citywide averages mask significant spatial variation. Long-term rents range from € 15.94/m²/month in peripheral zones to € 31.34/m²/month in the center, and short-term demand is equally heterogeneous. A flat in Duomo faces a completely different rental calculus than one in Baggio.

For a Milanese landlord, the question is simple: *will I earn more on Airbnb than with a long-term tenant?* The answer depends on location, local short-term demand, and what long-term rents look like in that specific area.

2.2 Motivation

Most existing work on Airbnb's impact in Milan looks at big-picture questions: regulation, tourism impact, or housing affordability at the city level. Useful for policy, but not if you're a landlord trying to decide what to do with a specific flat in Baggio. Neighbourhood-by-neighbourhood comparisons of short-term rental revenue against long-term rent, using actual listing data, are hard to find. Property owners need decision-support tools that reflect their specific location, not citywide averages.

Furthermore, most research focuses on the obvious high-rent tourist areas, paying less attention to peripheral neighborhoods where lower baseline rents might create arbitrage opportunities even with modest Airbnb demand. We wanted to look at the less obvious places: peripheral neighborhoods where lower baseline rents might turn even modest Airbnb demand into a worthwhile premium. So we computed a granular premium metric and built an interactive tool that landlords could actually use.

This project has two integrated parts. The analytical side answers a market research question: ***Where are short-term rental premiums highest, and what neighborhood characteristics appear to be associated with them?*** The dashboard side turns that research into something a landlord can actually use.

2.3 Research question

Which Milan neighbourhoods generate the highest short-term rental premium over long-term rent, and what may help account for variation in this premium across neighbourhoods?

We operationalise this through a premium metric comparing median Airbnb monthly revenue with long-term rent for a standardised 60 m² apartment, computed for each of Milan's 87 neighbourhoods.

3. Data and methodology

3.1 Data sources

Four datasets underpin the analysis:

Source	File	Description	Size
Inside Airbnb	listings.csv	Individual Airbnb listings in Milan	22,948 rows
Inside Airbnb	neighbourhoods.geojson	Neighbourhood boundary polygons	88 features
Immobiliare.it	milan-rent-by-zone.csv	Long-term asking rent by macro-zone (€/m ² /month)	31 zones
Manual	neighbourhood_to_zone_map.csv	Mapping from 87 Airbnb neighbourhoods to 31 rent zones	87 rows

Inside Airbnb is a public repository of scraped Airbnb data. The Milan dataset includes listing price, location, room type, availability, review counts, and neighbourhood assignment. The GeoJSON file provides polygon boundaries for map rendering.

Immobiliare.it is Italy's largest real estate portal. We collected zone-level asking rents (EUR per square metre per month) for 31 macro-zones in Milan. Each zone's data has its own reference month, because Immobiliare.it updates zones on independent schedules.

Because Airbnb uses 87 fine-grained neighbourhoods while Immobiliare.it uses 31 broader zones, we built a manual mapping file linking each Airbnb neighbourhood to its corresponding rent zone. All 87 neighbourhoods were matched.

3.2 Data cleaning

The table below summarizes the data cleaning pipeline for Airbnb listings:

Step	Records Before	Records After	Action Taken
Raw data	22,948	-	Initial dataset
Price conversion	22,948	20,553	Dropped 2,395 (10.4%) with missing prices
Outlier handling	20,553	20,553	Flagged 208 at p99 (€1,200/night) but retained
Final dataset	-	20,553	Used for neighbourhood-level analysis

Airbnb listings. The price was stored as a string (“\$125.00”), so we stripped the dollar sign and comma, then converted to float. 2,395 rows (10.4%) had missing prices and were dropped, reducing the dataset from 22,948 to 20,553 rows. We flagged 208 outliers at the 99th percentile (€ 1,200/night) but kept them in the dataset. Neighbourhood names were title-cased for consistency. Missing reviews_per_month values (2,896 remaining after the price-based drop) were filled with zero.

Long-term rent. We standardised zone names, parsed reference months into YYYY-MM format, and confirmed no duplicates. All 31 zones were retained. Rents ranged from € 15.94/m²/month (Bisceglie Baggio Olmi) to € 31.34/m²/month (Centro), with a median of € 20.31/m²/month.

Consistent time window. Immobiliare.it updates each zone on its own schedule, so reference months in the raw data range from October 2023 to February 2026. To make the data comparable across zones, we defined a consistent window of June 2025 to March 2026. This initially excluded 26 zones whose reference months were too old. We manually checked the 8 most outdated zone pages on Immobiliare.it, found that all had been updated to March 2026, and refreshed the raw data file. After rebuilding the pipeline, all 31 zones fell within the consistent window: 87 out of 87 neighbourhoods covered.

3.3 Premium calculation

We estimated Airbnb monthly revenue using the availability proxy:

$$\text{Estimated monthly revenue} = \text{price per night} \times (\text{availability_365} / 12)$$

availability_365 records how many days a listing is available for booking in the next year. It is an imperfect occupancy proxy: a listing available 365 days might be newly

posted rather than fully booked, and one available 0 days might be blocked by the host rather than sold out. We used median revenue at the neighbourhood level to limit the effect of outliers.

Long-term monthly rent for a standard apartment:

$$\text{Long-term monthly rent (60 m}^2\text{)} = \text{rent_€_m}^2\text{_month} \times 60$$

The premium metric:

$$\text{Premium (median, 60 m}^2\text{)} = (\text{median monthly revenue} - \text{long-term monthly rent 60 m}^2) / \text{long-term monthly rent 60 m}^2$$

A premium of +100% means the median Airbnb listing in that neighbourhood earns twice what a long-term lease on a 60 m² apartment would generate. Neighbourhoods with fewer than 30 Airbnb listings were flagged as `low_sample` and excluded from the main analysis.

3.4 Role of Claude Code

Claude Code handled the technical execution at every stage. It wrote the Python scripts for data cleaning (price conversion, missing-value diagnostics, outlier detection, neighbourhood-level aggregation), built the merge logic between the Airbnb and Immobiliare.it datasets using our manual mapping file, and generated all three visualisations (matplotlib bar chart, folium choropleth, plotly scatter plot).

We used Claude Code's plan mode extensively, which turns Claude into a collaborative agent rather than a one-shot code generator. Before each major step, we entered plan mode to discuss the approach: how to handle missing prices and outliers during data cleaning, which chart types and color scales to use for each visualization, and how to structure the dashboard layout and interaction logic. Plan mode let us review Claude's proposed strategy, ask questions, and agree on an approach before any code was written. This back-and-forth was especially useful for the dashboard, where design decisions (two-tab layout, cross-filtering, and apartment-size slider) benefited from iterative discussion rather than a single prompt.

Claude also developed the interactive dashboard: a single-page HTML/JavaScript application using Plotly.js, with cross-filtering between panels, dynamic premium recalculation tied to an apartment-size slider, a three-language toggle (EN/IT/CN), and a two-tab layout.

Debugging was a recurring use case. When 27 neighborhoods appeared grey on the initial choropleth, Claude helped trace the problem to stale rent data being excluded by the consistent-window filter. After we manually refreshed the source data, Claude rebuilt the full pipeline.

Our CLAUDE.md file gave Claude project-specific instructions: report currency in EUR, flag assumptions, use the YlGnBu color palette, prefer median over mean, and exclude low-sample neighborhoods from main conclusions.

4. Findings

4.1 Neighbourhood ranking by premium

The table below shows the top 10 neighbourhoods by short-term rental premium (excluding low-sample areas with fewer than 30 listings; consistent window June 2025 to March 2026):

Rank	Neighbourhood	Median Airbnb Revenue (EUR/month)	Long-Term Rent 60 m ² (EUR/month)	Premium
1	Duomo	4,542	1,880	+141.5%
2	Vigentina	2,650	1,194	+121.9%
3	Xxii Marzo	2,390	1,209	+97.7%
4	S. Siro	2,124	1,081	+96.5%
5	Brera	3,588	1,880	+90.8%
6	Centrale	2,400	1,405	+70.8%
7	Mecenate	1,740	1,022	+70.2%
8	Greco	1,934	1,145	+68.9%
9	Rogoredo	1,925	1,144	+68.3%
10	Baggio	1,587	956	+65.9%

Note: Long-term rent is asking price multiplied by 60 m².

Duomo's +141.5% premium comes from two sides: very high Airbnb demand (1,468 listings, median revenue €4,542/month) and the highest long-term rents in Milan (€ 31.34/m²/month in the Centro zone). But the ranking is not just a list of central tourist areas. S. Siro (rank 4, +96.5%) is peripheral, with long-term rents roughly half those of Centro. Baggio (rank 10, +65.9%) sits in one of the cheapest rent zones at € 15.94/m²/month. Short-term rental arbitrage is not exclusive to the city centre.

Figure 1 shows the top 15 neighbourhoods as a horizontal bar chart. Premiums range from Portello (+62.7%) to Duomo (+141.5%), spread across different parts of the city.

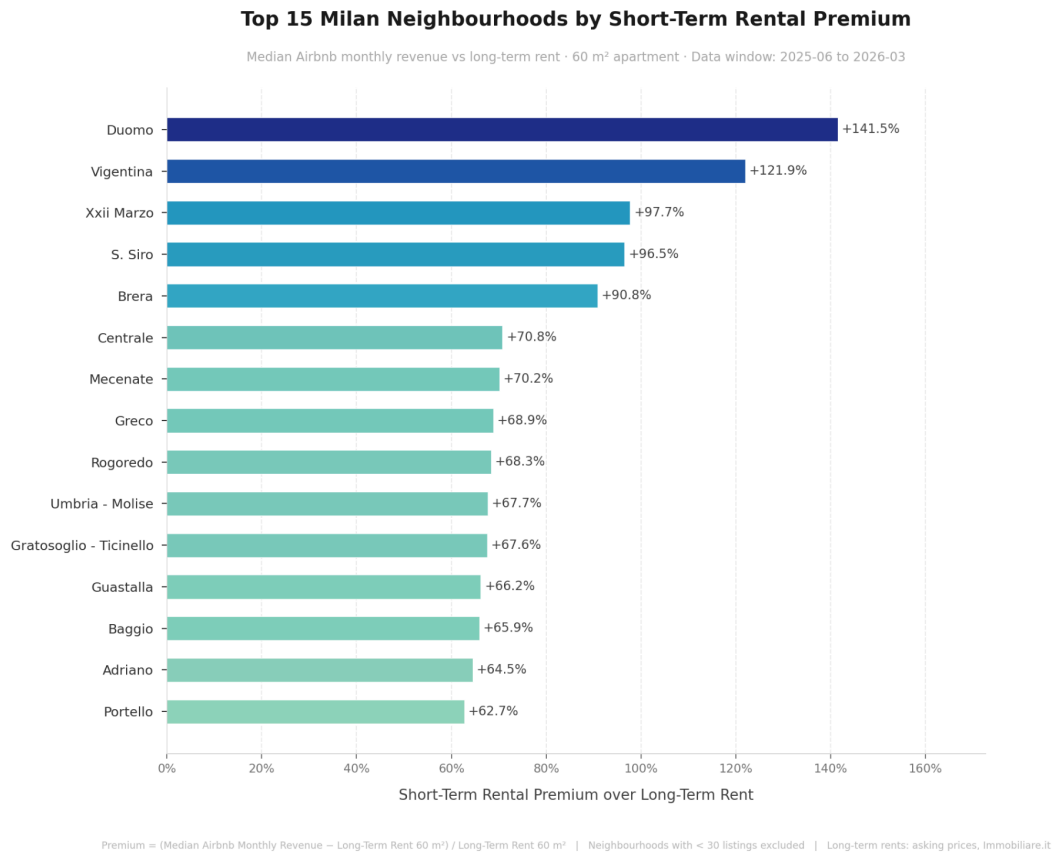


Figure 1: Top 15 Milan neighbourhoods by short-term rental premium.

4.2 Geographic distribution

The interactive choropleth map (Figure 2) colours each neighbourhood by premium level. High-premium areas are scattered across the city rather than concentrated in a central cluster. Duomo and Brera show the darkest shading, but so do pockets in the west (S. Siro), south (Vigentina, Rogoredo), and east (Xxii Marzo, Mecenate).

Only one polygon remains grey on the final map: Cascina Triulza - Expo, which has zero Airbnb listings. The other 87 neighbourhoods all have computed premiums. The 22 neighbourhoods with fewer than 30 listings can be toggled on or off in the dashboard's Deep Dive tab.

Short-Term Rental Premium by Milan Neighbourhood

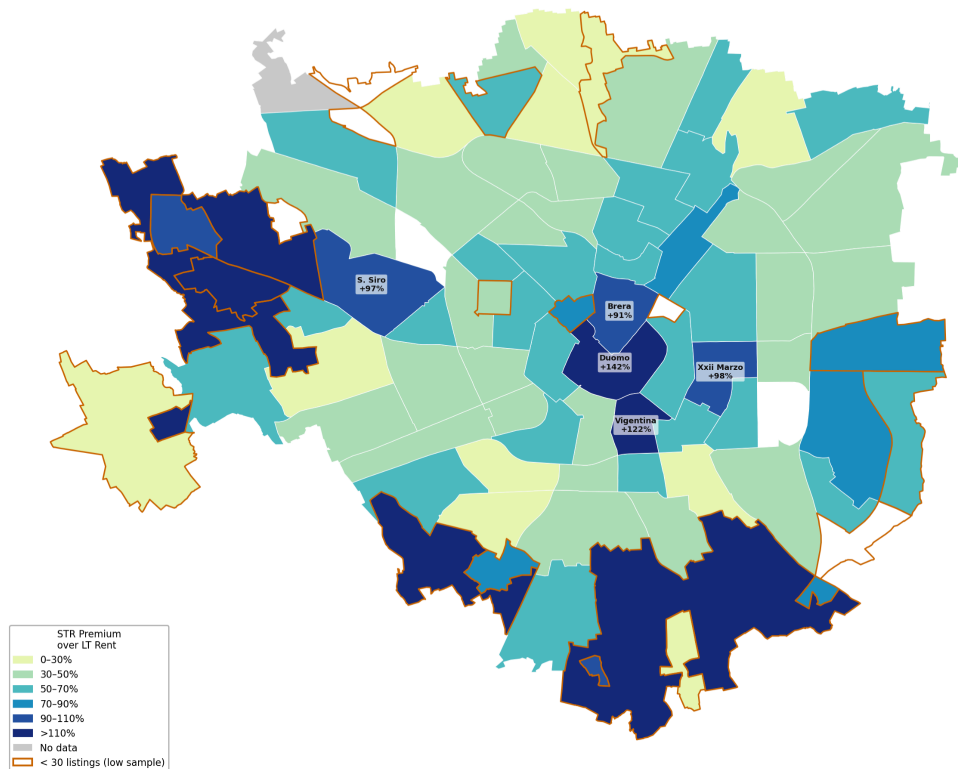


Figure 2: Choropleth map of STR premium by neighbourhood.

Interactive version: visualizations/choropleth_premium_median_60m2.html

The geographic pattern matters for landlords outside the centre. Low long-term rents in peripheral zones set a lower baseline, so even modest Airbnb revenue can produce a high percentage premium. A landlord in Baggio does not need to match Duomo-level nightly rates to see a worthwhile premium over their local long-term rent.

4.3 Rent level is a weak predictor of premium

The scatter plot (Figure 3) places each neighbourhood's long-term rent on the x-axis and its premium on the y-axis, with marker size proportional to listing count and colour by rent quartile. The OLS trend line has a weak positive slope ($R^2 = 0.05$, $p = 0.078$). Rent level explains about 5% of the variation in premium.

You might expect expensive areas to also have the highest Airbnb premiums. That is only partly true. Duomo has both the highest rent and the highest premium, but Vigentina, at a moderate rent level (€ 19.90/m²/month), is second with +121.9%. These suggest two different paths to a high premium: being in a high-demand tourist area where Airbnb prices are very high, or being in a cheaper area where even moderate Airbnb revenue represents a large percentage gain over the local rent.

On the other end, several high-rent neighbourhoods have premiums below the median. Some areas in the Quadronno Palestro Guastalla and Garibaldi zones show premiums around +30-50%. High long-term rents set a high bar, and short-term rental revenue does not always clear it by much.

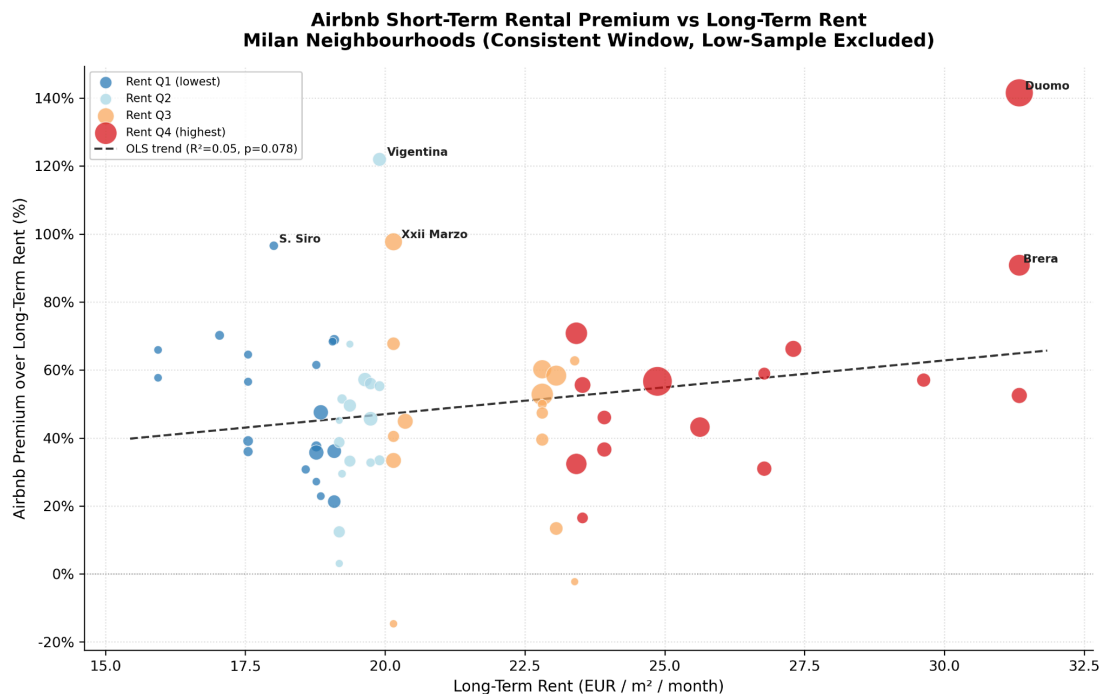


Figure 3: Scatter plot of STR premium vs long-term rent.

[4.4 Interactive dashboard \(Click the link to open the dashboard\)](#)

4.4.1 User-Centered Design & Logic

We built the dashboard for landlords, not researchers. The goal was to answer the following: Should I switch to Airbnb? for a specific flat, not to display aggregate patterns? Claude Code wrote the code; we argued about the design. Claude Code handled the technical implementation; our contribution was the design specification, user framing, and iterative direction through prompts.

We anchored every decision to a target user persona: Marco, a 58-year-old Milanese landlord who opens the dashboard on a tablet with one question, "Vale la pena?" (Is it worth it?). To test an early version, we prompted Claude to roleplay as Marco and give candid feedback. This exercise revealed a fundamental mismatch: the dashboard had been organized around academic metrics (premium percentages and scatter plots with regression lines), while a property owner thinks in monthly euro figures and personal decisions. The exercise redirected the entire design.

The biggest structural decision was the two-tab layout, based on progressive disclosure: Show the simplest useful view first and let users opt into complexity. A landlord opening the page for the first time should see a map and a single call-to-action, not a scatter plot. The Overview tab holds the map, a neighborhood detail card, and three headline statistics; the Deep Dive tab holds the bar chart, scatter plot, and technical footnotes.

We also rewrote all user-facing labels to be conversational. "Where is your flat?" replaced "Select neighborhood"; "How big is your flat?" replaced a labeled slider. This framing was specified directly in our prompts, with an explicit instruction to use dialogue-like language throughout the overview.

One iteration illustrates the design process. An early version exposed a checkbox labelled "Only show areas we're confident about." Rather than asking for a relabeling, we wrote a prompt asking for root-cause analysis: Why did this checkbox exist? What problem created it? And how should it actually be solved? The discussion surfaced that the checkbox was leaking an internal data-quality concern into a screen designed for property owners—a data scientist's solution misapplied to a user interface. The fix: exclude low-confidence areas by default, explain it in the Deep Dive tab, remove the checkbox from Marco's screen entirely.

4.4.2 Implementation & Accessibility

Accessibility and mobile access were shaped by thinking through Marco's real situation. We added a three-level font-size toggle because our target user might be an older Italian landlord reading on a tablet. For mobile access, we first asked Claude to assess feasibility before building anything, which correctly identified the key constraint (a QR code works only over HTTP, not as a local file) and led to a public Netlify deployment as the clean solution. The dashboard supports English, Italian, and Chinese; an apartment-size slider (40 - 120 m²) recalculates all figures in real time; and clicking any neighborhood on the map cross-filters the bar chart and scatter plot. Each of these features was introduced through a dedicated prompt specifying the requirement, the user rationale, and the intended placement.

5. Discussion and recommendations

Our research question asked, "Which **Milan neighborhoods generate the highest short-term rental premium over long-term rent, and what may help account for variation in this premium across neighborhoods?**"

The answer is twofold. First, **Duomo ranks highest** at +141.5%, but premiums are geographically widespread, including peripheral areas like S. Siro (+96.5%, rank 4) and Baggio (+65.9%, rank 10). Second, **rent level is a weak predictor** of premium variation ($R^2 = 0.05$, $p = 0.078$), suggesting that localized factors such as tourism

demand, accessibility, business travel patterns, and transport links matter more than baseline rent levels. This means landlords cannot rely on citywide generalizations or assumptions about centrality; neighbourhood-specific conditions drive profitability.

5.1 Interpreting the premium

The premium numbers are gross revenue, not what you actually pocket. Short-term rental carries costs that long-term leasing does not: cleaning between guests, furnishing, Airbnb's host commission (~3%), higher tenant turnover, and the landlord's own time. A premium of +60% means estimated gross Airbnb revenue exceeds the long-term asking rent by 60%; it does not mean the landlord pockets 60% more. The actual net advantage after operating costs and vacancy would be smaller.

Still, premiums in the +60% to +140% range leave considerable room for costs before the comparison flips. In some top-performing neighborhoods like Duomo, Vigentina, or S. Siro, Airbnb may still remain more attractive than a long-term lease after reasonable operating deductions, but this cannot be confirmed without net-income modeling.

5.2 What this means for landlords

The analysis does not support the view that Airbnb's advantage is confined to the tourist center. High premiums appear in Duomo and Brera, yes, but also in S. Siro, Baggio, and Rogoredo. These peripheral areas share some source of short-term demand, whether from business travel, trade fair visitors, or transport links, even if they do not attract the same volume of tourists as the centre.

The variation across neighbourhoods is also larger than we expected. Some high-rent areas show premiums below +40%; some low-rent areas exceed +90%. A landlord evaluating the switch to Airbnb should look at their own neighbourhood specifically, not at Milan-wide statistics.

That said, the dashboard is a starting point. Individual decisions depend on things our data does not capture: property condition, building-level rental restrictions, seasonal demand patterns, and how much work the owner is willing to put into managing guests.

5.3 Policy implications

Short-term rental premiums are geographically widespread. They are not concentrated in a tourist corridor around the Duomo. The financial incentive to convert a property from long-term to short-term rental exists across the city, including in peripheral

zones. Regulatory approaches that focus only on central areas may underestimate the scope of the short-term rental market.

6. Validation and limitations

6.1 How we verified Claude's outputs

To ensure the reliability of the dashboard, we did not take the AI-generated outputs at face value. Our validation process followed two steps: technical verification and real-world user testing.

Technical Verification

We first performed manual spot-checks by extracting raw data from the CSV files for specific districts, including **Duomo, Vigentina, and S. Siro**. By manually calculating the premiums for these areas, we confirmed that the dashboard's computations were accurate. During testing, we identified an issue where 27 neighborhoods appeared as "no data" (grey) on the map. Through manual cross-referencing of the **join keys**, we traced this to an overly restrictive data filter that excluded older records. We corrected this by adjusting the filter and adding diagnostic prompts to flag such "silent errors" in the future. Additionally, we addressed a statistical discrepancy: an early version of the dashboard used the **mean (average)**, which caused inflated figures for areas with very few listings, such as **Parco Dei Navigli**. Switching the primary metric to the **median** resolved these outliers and ensured more stable results.

User Acceptance Testing

Once the technical logic was verified, we conducted trial sessions with **local property owners** in Milan to evaluate the tool's practical utility. The feedback was overwhelmingly positive; participants reported that the interface was intuitive and easy to navigate. This confirmed that our earlier design choices—such as prioritizing the "Overview" tab with direct monthly revenue figures and relocating complex academic charts to a "Deep Dive" section—were effective. The testing proved that the dashboard functions successfully as a decision-support tool for its intended audience.

6.2 Data limitations

Asking prices, not transaction prices. Immobiliare.it reports asking rents, which may differ from what tenants actually pay. Asking prices tend to be higher, so our long-term rent figures may slightly overstate actual rents, which would understate the true premium.

Availability as an occupancy proxy. `availability_365` records how many days a listing is open for booking, not how many days are booked. A listing with high availability might be underperforming; one with low availability might be blocked by the host rather than sold out. This is a standard proxy in Airbnb research, but it is noisy.

60 m² assumption. We needed an apartment size to convert per-square-meter rents into monthly rent. 60 m² is a reasonable middle ground for Milan, but apartments vary. The dashboard lets users adjust this between 40 and 120 m².

Geographic granularity mismatch. The 87-to-31 mapping between Airbnb neighborhoods and Immobiliare.it zones is an approximation. Some neighborhoods assigned to the same zone probably have different actual rent levels.

Single snapshot. The Airbnb data is a single point in time. We do not capture seasonal variation, which could matter in a tourism-driven market.

Gross, not net. The premium compares gross revenue. It does not account for Airbnb fees, cleaning costs, taxes, or management time.

6.3 Errors caught and corrected

Silent join failure. The initial choropleth looked plausible (27 grey polygons) and raised no errors. The problem turned out not to be a code bug but a data issue: stale rent records excluded by the time-window filter. We added explicit join diagnostics (matched vs unmatched feature counts) so this type of silent failure would be caught immediately.

Mean vs. median. An early version used mean revenue, which produced misleading results for neighborhoods with extreme outliers. Parco Dei Navigli had a mean of € 41,694/month and a median of € 3,570/month, based on just 7 listings. Switching to median fixed this.

Build script bug: `build_dashboard.py` had two bugs: it re-read the template file after replacing the GeoJSON placeholder (overwriting the replacement), and it never actually wrote the output file. Every previous session had been opening the unmodified template without realizing it. We caught this during dashboard polishing.

6.4 What we would do differently

With more time, we would replace `availability_365` with actual booking data or review frequency as a more reliable occupancy proxy. We would also add an operating-cost layer to the dashboard (Airbnb fees, cleaning, furniture depreciation) so that users see net income rather than gross revenue. Finally, collecting multiple snapshots of Airbnb

data across different months would let us account for seasonality, which the current single-snapshot approach misses.

7. References

Data sources

1. Inside Airbnb. Milan listing data and neighbourhood GeoJSON boundaries. Retrieved March 2026 from <http://insideairbnb.com/get-the-data/>
2. Immobiliare.it. Long-term asking rent data by zone (€/m²/month) for 31 Milan macro-zones. Retrieved March-April 2026 from <https://www.immobiliare.it/en/mercato-immobiliare/lombardia/milano/>

Tools

1. Claude Code (Anthropic). Primary tool for data cleaning, analysis, visualisation, dashboard development, and report drafting.
2. Python libraries used: pandas, matplotlib, folium, plotly, geopandas, adjustText, scipy.
3. Plotly.js. Client-side JavaScript library for interactive dashboard visualisations.
4. Netlify. Static site hosting for dashboard deployment.